

Early experience affects foraging behavior of wild fruit-bats more than their original behavioral predispositions

Adi Rachum, Lee Harten, Reut Assa, Aya Goldshtein, Xing Chen, Nesim Gonceer, Yossi Yovel 

School of Zoology, Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel • Sagol School of Neuroscience, Tel Aviv University, Tel Aviv, Israel

 https://en.wikipedia.org/wiki/Open_access Copyright information

eLife Assessment

This paper provides **important** insight into how early life experience shapes adult behavior in fruit bats. The authors raised juvenile bats either in an impoverished or enriched environment and studied their foraging behaviors. The evidence is **convincing** that bats raised in enriched environments are more active, bold, and exploratory. The work will be of interest to ethologists and developmental psychologists.

<https://doi.org/10.7554/eLife.103220.2.sa2>

Abstract

There are immense consistent inter-individual differences in animal behavior. While many studies have documented such behavioral differences, often referred to as individual personalities, little research has focused on the underlying causes and on determining whether they are innate or based on individual experience. Moreover, most studies on animal personalities have described consistent differences in behavior under laboratory conditions. We aimed to examine the impact of the early experienced environment on individual animal behavior, and to compare it to that of the individual's original genetic predisposition. Additionally, we explored the correlation between personality traits measured indoors and the animal's outdoor behavior. We studied Egyptian fruit bats, in which vast behavioral variability and plasticity have already been demonstrated. We raised bats in a captive colony under either enriched or impoverished environments and assessed their personality under controlled laboratory conditions. We then released the bats into the wild and tracked their foraging using GPS. Bats that had experienced an enriched environment during early life displayed increasing boldness and exploratory behavior when foraging outdoors, demonstrating how early-life experience can affect adult behavior. The individuals' original predispositions did not predict their later foraging behavior. Our findings shed new light on the interplay between innate and experienced-based effects on individual behavior.

Significance statement

Animals exhibit individual personalities, but how they are shaped is unknown. By employing controlled manipulations on new-born bat pups and GPS-tracking them outdoors as adults, we reveal that the early life conditions bats are exposed to have a significant impact on their foraging behavior as adults. We moreover show that the original individual predispositions of the bats do not predict their behavior. These findings enhance our understanding of the developmental factors that shape animal behavior and emphasize the vital importance of environmental enrichment during early life stages.

Introduction

Anyone who has ever observed animals knows that individuals persistently differ in their behavior. These behavioral tendencies, enable distinguishing individuals from one another and remain consistent over time and across various situations, have been termed “animal personalities”¹. While some opposition exists regarding the use of this term², it is nonetheless commonly employed^{3,4}, and we use it here to describe a set of behavioral traits that remain consistent within individuals over time and across contexts¹.

Personality traits have been extensively documented in various taxa^{1,5,6}. They are typically assessed through the measurement of proxies for behavioral traits such as boldness, activity, and exploration⁷. However, while individual personality differences have been observed in numerous species^{5,6,8,9}, only a limited number of studies have investigated the underlying factors that contribute to their emergence.

Broadly speaking, the ontogeny of an animal's personality can be affected by its genetic predisposition and by its individual experience. To date, only a few studies have explored how individual personality develops, and how the interplay between genetic factors, maternal effects, and early experiences or environmental effects shape personality^{1,6,10–12}. In the few studies that did so, the effects of these factors have been observed exclusively in captive settings^{12–16}. For instance, researchers found differences in personality traits, such as boldness and exploration, in guinea pigs raised in captivity under different photoperiods simulating spring or autumn conditions¹¹. Similarly, intertidal gobies reared in different habitat complexities have shown differences in spatial learning laboratory experiments: gobies that developed in complex rock pools demonstrated faster learning abilities compared to those raised in homogenous sandy shores¹⁵.

Only a handful of studies have examined behavioral traits under both captive and natural conditions. One such experiment showed that the foraging behavior of birds in captivity was consistent with their behavior in the field. For example, the exploratory tendencies in captivity were correlated with seeking new feeding sites in the wild¹⁷. There is also a scarcity of hypothesis-driven experiments that employ specific manipulations to investigate the ontogeny of personality. In this study, we aimed to elucidate the roles of genetic predisposition and early life experience on free-foraging fruit-bat personality using a controlled manipulation.

Egyptian fruit bats have been shown to exhibit immense individual differences both in indoor personality tests¹⁸ and when foraging in the wild¹⁹. These bats have also been shown to exhibit environmentally dependent behavioral plasticity. Specifically, the same individual fruit-bats were much more exploratory when foraging in urban environments than when foraging in

rural environments ¹⁰[10](#),²⁰[20](#). Moreover, fruit-bats that roost in cities were shown to be even more exploratory than rural fruit-bats that commute into cities, suggesting that the environment might have an additive effect on their behavior.

In this study, we performed a controlled manipulation to determine whether the environment that young bat pups experience can affect their personality and outdoors foraging. Specifically, we examined whether exposure to an enriched environment as pups, affects the bats' behavior as measured in the lab and when they later forage in the wild as adults. Living in enriched or dynamically changing environments has been linked in multiple species to many behaviors, including increased risk-taking and boldness ¹⁰[10](#),²¹[21](#),²²[22](#). Maturation in an enriched environment can also affect learning capabilities, as noted above regarding gobies ¹⁵[15](#), and motor performance as demonstrated in laboratory rats ¹³[13](#).

In our study, we manipulated the extent of environmental enrichment experienced by juvenile fruit-bats in a captive environment and examined the correlation between this early experience and their foraging behavior in the wild several months later. We compared the effects of experience to the bats' original behavioral predisposition, estimated when they were naïve pups (~4 months old; prior to any manipulation). We also examined personality traits over a prolonged period to confirm their individual behavioral consistency, and we explored the relationship between the personality traits observed in captivity and the outdoor foraging behavior. We hypothesized that both early-life experience and original predisposition would interact in shaping individual personalities.

Results

Assessing Personality in the lab

To assess bats' personality, we ran young bats through the multiple-foraging box paradigm, which was developed in our lab and has been shown to measure behavioral variability along multiple axes ¹⁰[10](#). A total of 40 bats participated in the foraging box experiments, which were repeated a maximum of five times per individual over a period of almost five months (**Figure 1A** [10](#), see Methods). The foraging box experimental setup was designed to assess the innate personality traits in bats which we could then use as parameters when trying to explain their foraging outdoors. Repeated measurements were used to confirm that the traits met the criteria for personality, such as consistency over time. The experiment comprised baseline trials for each bat on two consecutive nights, upon the bats reaching an average age of 4.1 ± 1.4 months (trials 1-2; Mean $\pm$ STD). Subsequently, we randomly divided the bats into two colonies, each exposed to different environmental conditions. Both colonies were provided with the same basic diet sufficient for all bats in the colony. Importantly, bats in both colonies were raised in large social groups of similar size. The **enriched colony** underwent frequent changes in their surroundings for a period of ~10 weeks, through the introduction of various enrichments that forced the bats to practice trial and error to obtain (Methods). While the **control impoverished colony** was exposed to a stable environment with minimal changes. A post-enrichment boxes trial was performed immediately after this period (Trial 3, on average 72 days after the previous). Finally, two post-release boxes trials were performed after the bats were released into our open colony and could freely forage in the wild and explore their natural environment (Trials 4-5, average 45 and 58 days after their release into the wild). Post-enrichment and post-release trials were conducted to assess whether early life experiences had an effect on the development of bat personality and in order to identify persistent personality traits that can be used as a proxy of the individual innate predispositions. We used these trials to assess proxies of the three above-noted behavioral traits: boldness, exploration, and activity. To assess the behavioral traits during the foraging box experiments, we scored the bats' actions. Specifically, we recorded how many times they

approached and took food from a foraging box, the number of different boxes they explored (i.e., landed on), and the total number of actions they performed during the night (**Figure 1A** [↗](#) and see Methods).

When plotting the three behavioral traits (boldness, exploration, and activity) in 3D they seem to form a triangle (**Figure 1B** [↗](#)) – reminiscent of a pareto front ²³[↗](#) – suggesting a trade-off between the different traits. The three vertices of the triangle (often referred to as the archetypes when analyzing a pareto front) represent individuals that exhibit: (a) high exploration, high activity, and medium-low boldness. (b) high boldness, low activity, and low exploration, and (c) low boldness, low activity, and low exploration (very few bats). These three archetypes suggest a trade-off between boldness on the one hand, and activity and exploration on the other hand.

For all three behavioral traits, we found a significant positive correlation between baseline (Trial 1) and the post-enrichment trial (Trial 3) examined on average 72 days apart. A significant positive correlation was also found between bats' boldness during the baseline (Trial 1) and the second post-release trial (Trial 5), examined on average 144 days apart. These findings suggest that these behavioral traits, and especially boldness, remain consistent at least throughout the bat's early life (**Figure 1** [↗](#) C1-3, Pearson correlation test: $r=0.66$ $P<9.7e-0.5$, $r=0.66$ $P<9.1e-0.5$, and $r=0.51$ $P<0.004$; for Trials 1-3 for boldness, exploration and activity respectively, and $r=0.6$ $P<0.021$, for Trials 1-5 boldness, **Table 1** [↗](#)). These results did not change also when examining the enriched and impoverished colonies separately. To compute an integrative measurement of personality, we ran a PCA analysis on the data from trials 1 and 5. We found a significant Pearson correlation ($P=0.03$, $r=0.58$, **Supplementary figure S1** [↗](#)) between the individuals' scores on PC1 in trials 1 and 5 suggesting that this measure captures a persistent representation of behavior that takes both boldness and exploration into account (the activity weight was very low).

Notably all three personality traits exhibited significant correlations in the two baseline trials suggesting that indeed they capture behavioral tendencies (**Figure 2** [↗](#)).

The temporal dynamics of boldness and exploration revealed similar patterns: both increased between the first two consecutive baseline trials (**Figure 2** [↗](#)). Following the enrichment condition (i.e., in Trial 3) they decreased back to the level of the first baseline and then increased again in the two post-release trials (4-5). Boldness was significantly higher in the last trial compared to in the first one ($P < 1.9e-07$, see discussion). A mixed-effect Generalized Linear Model (GLMM) was used, with the behavioral traits set as the response variable, trial number (1 or 5) as a fixed factor, and bat ID as a random effect (see **Table 2** [↗](#)). While exploration showed a fairly similar pattern. Activity, in contrast, showed a different pattern. It decreased continuously throughout the trials and was significantly lower in the last trial compared to in the first one ($P < 0.0001$, GLMM as above). An individual-level comparison between the first and last trial is presented in **Figure 2C** [↗](#)

Despite the above-reported changes in behavior over time, the environment that the animals experienced as juveniles (enriched or impoverished) did not have a significant effect on their personality as measured in the lab (GLMM - with the behavioral traits set as the response parameter, the trial number and the interaction between trial number and the environmental condition as fixed factors, and bat ID as a random effect). The interaction between trial and environmental condition was not significant (**Table 3** [↗](#)). When examining only Trials 4-5, i.e., following their release into nature, the enriched bats were observed to be significantly bolder than the non-enriched bats ($P=0.003$, GLMM as above). However, this seems like a result of post-selection resulting from which bats remained in our colony and could be tested (see Discussion).

Figure 1.

Assessing personality in the laboratory.

(A) Schematic illustration of the experiments. (B) A 3D plot of the three behavioral traits, estimated during Trial 1. (C1-3) Behavioral traits were consistent over time. The behavioral traits of the first and third (post-enrichment) trial are presented. Positive correlations were found for all three behavioral traits over a period of more than 10 weeks. For the enriched (blue lines), the impoverished (orange lines), and both environments together (not shown), respectively: (C1) Boldness; (C2) exploration; and (C3) activity. (D) Boldness of the first and fifth (post-release 2) trial. Dashed line represents the Y=X line.

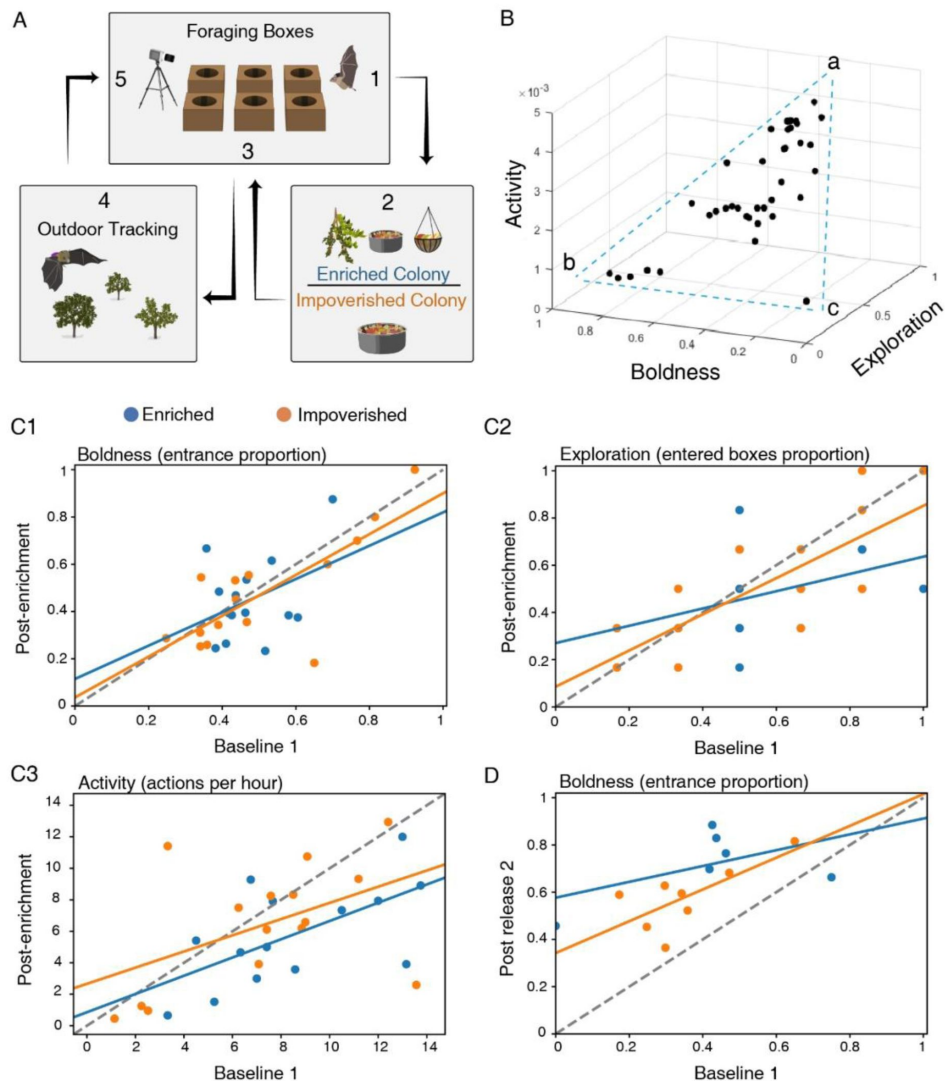


Table 1.

Pearson's correlation test.

	Boldness 1-3	Exploration 1-3	Activity 1-3	Boldness 1-5
Pearson r value	0.66	0.66	0.51	0.6
Pearson p value	<9.7e-05	9.1e-05	0.004	0.021

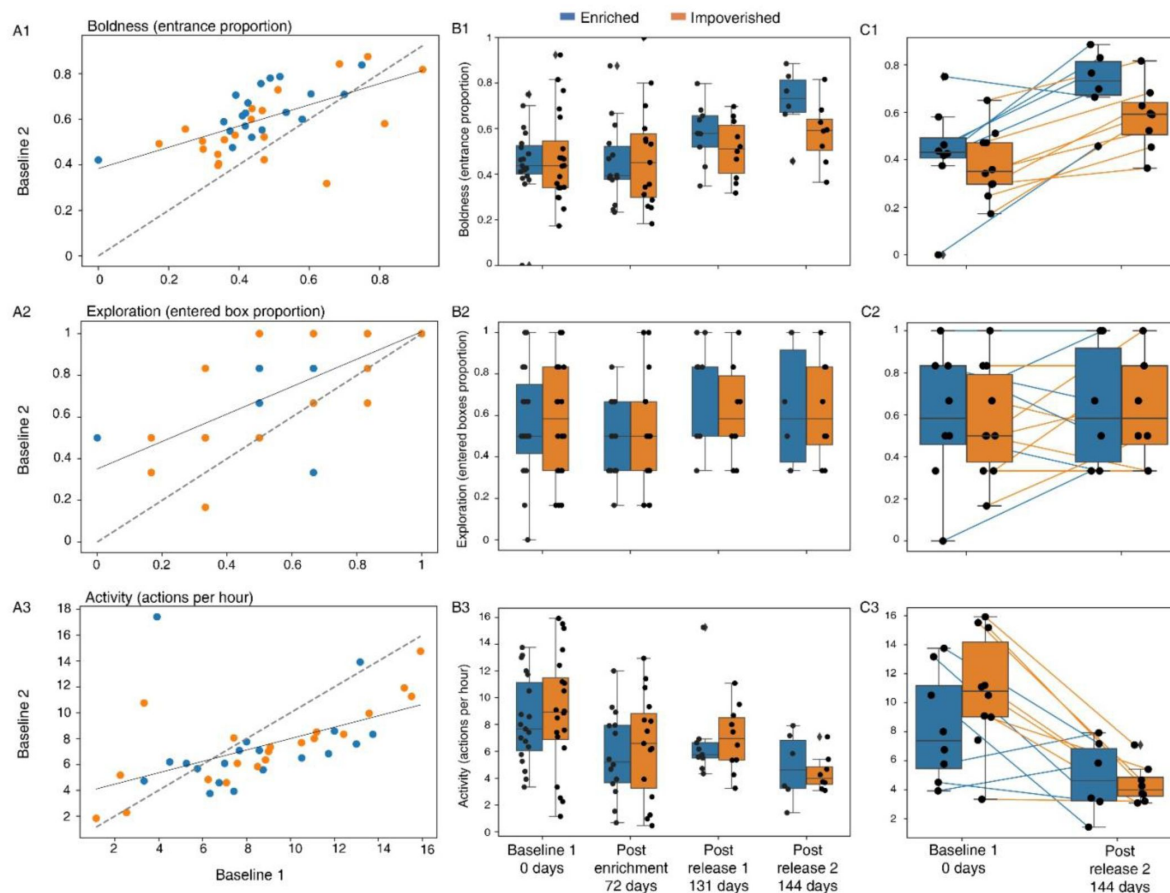


Figure 2.

Personality measurements over time.

(A) The behavioral traits of the first and second baseline trials, performed in two consecutive nights, are presented. Positive correlations were found for all three behavioral. Boldness (Pearson $r=0.59$, $p=5.8e-05$) (A1); exploration (Pearson $r=0.70$, $p=5.3e-07$) (A2); and activity (Pearson $r=0.52$, $p=0.0005$) (A3). Dashed line represents the $Y=X$ line. (B) Box plots depicting the bats' behavior across the five trials according to three traits: Boldness (B1), exploration (B2), and activity levels (B3). (C) Paired box plots show individual bat values in the first trial (baseline 1) and the fifth trial (post-release 144 days apart on average) for each trait: Boldness (C1), exploration (C2), and activity levels (C3). Lines connect repeated measurements from the same individual, illustrating within-individual behavioral changes over time. The number of bats participated in each trial is given in Methods, Table 8. In the baseline trials, no significant differences were found between the two environmental conditions (enriched / impoverished) for any of the three behavioral traits we measured ($P = 0.62$, $P = 0.84$, $P = 0.72$ for boldness, exploration, and activity levels, respectively, $n=40$, Table S1). Figure S2 also shows the indication. The experimental phase and the time in days from the day of the first trial are depicted on the X-axis. Box plots show the 25% and 75% percentile. Medians and whiskers based on 1.5 IQR are shown.

Table 2.

mixed GLM results for the comparison between trials 1 and 5 response ~ trial + (1 | bat ID).

Response	AIC	BIC	LogLikelihood	Deviance				
Boldness	-36.379	-28.498	22.189	-44.379				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.408	0.031	13.16	51	5.03e-18	0.346	0.471
	Trial number	0.056	0.009	6.02	51	1.902e-07	0.037	0.075
Response	AIC	BIC	LogLikelihood	Deviance				
Exploration	12.074	19.956	-2.037	4.074				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.545	0.049	11.02	51	4.248e-15	0.446	0.644
	Trial number	0.018	0.016	1.157	51	0.252	-0.013	-0.013
Response	AIC	BIC	LogLikelihood	Deviance				
Activity	-580.34	-572.46	294.17	-588.34				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.002	0.0001	13.737	51	9.050e-19	0.002	0.003
	Trial number	-0.0002	7.135e-05	-4.065	51	0.0001	-0.0004	-0.0001

Response	AIC	BIC	LogLikelihood	Deviance				
Boldness	-100.5	-82.807	56.25	-112.5				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.448	0.0444	10.078	137	3.266e-1	0.36	0.536
	Environmental condition Impoverished	0.007	0.061	0.117	137	0.906	-0.115	0.129
	Trial number	0.038	0.014	2.744	137	0.006	0.010	0.066
	Environmental condition Impoverished:Trial	-0.011	0.019	-0.58	137	0.561	-0.049	0.026
Response	AIC	BIC	LogLikelihood	Deviance				
Exploration	-0.46128	17.231	6.230	-12.461				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.625	0.066	9.467	137	1.13e-16	0.494	0.755
	Environmental condition Impoverished	-0.010	0.092	-0.112	137	0.910	-0.192	0.171
	Trial number	-0.003	0.019	-0.182	137	0.855	-0.041	0.034
	Environmental condition Impoverished:Trial	0.0030.006	0.025	0.247	137	0.804	-0.043	0.054
Response	AIC	BIC	LogLikelihood	Deviance				
Activity	-1586	-1568.3	799.01	-1598				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.002	0.0002	11.017	137	1.319e-2	0.002	0.003
	Environmental condition Impoverished	0.0002	0.0003	0.908	137	0.365	-30e-5	0.0009
	Trial number	-0.0002	7.292e-05	-3.273	97	0.001	-30e-3	-9.e-05
	Environmental condition Impoverished:Trial	-9.2e-05	9.640e-05	-0.959	137	0.338	-20e-5	9.8e-05

Table 3.

**mixed GLM results for all tested trials (1-5) response
~ 1 + Trial*EnvironmentalCondition + (1 | bat ID).**

Outdoor foraging behavior

After releasing the bats into our open colony, in which they were free to fly out, we tracked them using GPS devices as they explored the world for the first time and foraged in their natural environment. We analyzed data from 19 bats, obtaining data from 39.2 ± 17.0 (Mean $\pm$ SD) foraging nights per bat on average, accounting for $72.5 \pm 8.3\%$ (Mean $\pm$ SD) of the individuals' foraging nights. We used the following movement parameters as representatives of the bats' outdoor behavior: the total nightly time spent foraging as a proxy for activity; the explored area as a proxy for exploration; and the maximal distance from the colony as a proxy for boldness (see Methods).

The enriched bats significantly differed from the impoverished bats in all these movement parameters (**Figure 3** and Figures S3-S4). Specifically, they spent significantly more time out foraging every night (3.5 ± 1.9 vs. 2.8 ± 1.8 hours), flew farther from the colony (1.3 ± 1.8 vs. 0.8 ± 1.13 km), and explored larger areas (7.82 ± 6.72 vs. 3.39 ± 3.86 km²); all values represent the Mean $\pm$ SD for the entire study period (19 bats). In this analysis we included all bats and averaged according to the number of nights we recorded per bat. Examining of the data after 15-20 nights outside where all bats have been out for a similar period ($n=17$ bats with data) the pattern was the same and the values were: 4.02 ± 1.19 vs. 2.8 ± 0.75 hours, 1.66 ± 1.77 vs. 0.6 ± 0.28 km, 2.34 ± 2.71 vs. 0.23 ± 0.23 km².

To assess the relative effects of experience (i.e., enriched/impoverished) on foraging and to compare it to the bats' original predisposition, we ran a GLMM, with their original predisposition (Trial 1) and environment (enriched / impoverished) as fixed factors. We ran this analysis two times using either boldness or the PC1 score that accounted for both boldness and exploration of Trial 1 as a proxy for the bats' innate predisposition, because these had been examined before the bats were exposed to the different environments.

The analysis revealed that the environment in which the bats were reared was more important than their original predisposition for explaining their outdoor foraging behavior. While the effect of predisposition was not significant, the enriched environment significantly correlated with spending more time outside, flying to a significantly larger maximal foraging distance, and exploring a larger area ($P < 0.028$, $P < 0.01$, and $P < 0.02$, respectively). GLMM was used (see **Table 4a-b**, with environmental condition as the predisposition and experience as fixed factors). A model that includes an interaction between the predisposed boldness, exploration, or activity and the environmental condition, showed a worse fit (higher AIC). The bats' experience (represented by the number of nights they spent foraging until the day of the assessment), also had a significant positive effect, increasing the nightly foraging time and the maximal distance ($P < 0.0006$, and $P < 0.004$; GLMM for the time spent outside and the maximal nightly distance respectively, **Table 4a-b**). The bats' age (days from birth) did not have a significant effect. However, the proportion of nights that bats spent foraging outside the roost was not significantly associated with environmental condition, nor was it influenced by sex or age ($P = 0.95$, see **Supplementary Table S2**).

We also tested the correlation between the behavioral traits measured in the wild and found a significant positive correlation only between the maximal distance and the area the bat explored, (Pearson correlation test; **Table 5**).

Lastly, we compared the foraging behavior outdoors to the behavioral personality traits measured in the indoor foraging task (assessed ~two weeks following their release into the open colony - Trial 4). There was no significant correlation between indoor personality and any of the outdoor foraging characteristics. (Pearson correlation test, $P > 0.05$ for all comparisons).

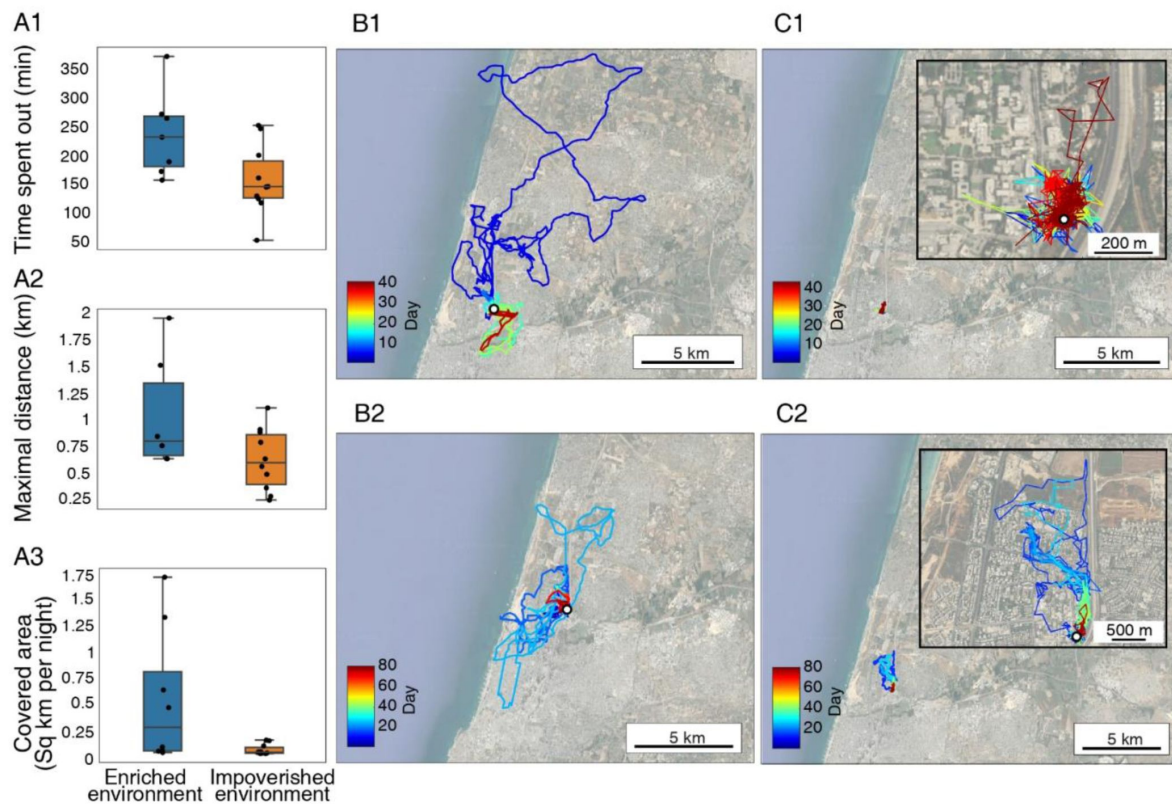


Figure 3.

Early environmental exposure affects outdoor foraging behavior.

(A1) Time spent by the bats outside the colony each night (in minutes). **(A2)** Distance to the furthest point from the colony per night. **(A3)** The area explored by the bats. The data for each bat were estimated for the period between its 15th and 20th day outdoors (n=17 bats). Median and whiskers based on 1.5 IQR are shown. **(B-C)** The complete movement of four individuals - two from each colony. **(B)** Individuals raised in the enriched colony, data shown for two individuals, each representing a different measured season (B1,B2); **(C)** Individuals raised in the impoverished colony, data shown for two individuals, each representing a different measured season (C1,C2). Colors depict time in days. The open colony is marked by a white circle. Insert in C1-2 zooms-in on the tracks. More examples of individual bat movement maps are presented in [Figure S3](#).

Table 4a.

GLMM results for outdoors measurements with the first boldness as predisposition.

The lines depicting results for the environmental and the predisposition effects are highlighted in green and blue respectively. response ~ 1 + Environmental_condition + Baseline_boldness + Sex + Age + Days_spent_outside + (1 | Bat_ID)

Response	AIC	BIC	Loglikelihood	Deviance				
Time spent outside	8638.2	8674.8	-4311.1	8622.2				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	185.42	74.081	2.469	710	0.013	38.014	332.83
	Environmental condition Impoverished	-57.736	26.374	-2.189	710	0.028	-109.52	-5.956
	Boldness of the first baseline	-32.783	78.474	-0.417	710	0.676	-186.85	121.29
	Bat sex Male	-9.0905	26.014	-0.349	710	0.726	-60.163	41.982
	Bat age	0.029	0.217	0.135	710	0.892	-0.397	0.456
	Number of days spent outside until assessment day	1.416	0.351	4.028	710	6.1e-05	0.726	2.106
Response	AIC	BIC	Loglikelihood	Deviance				
Maximum distance per night	12437	12474	-6210.7	12421				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	1543.2	651.67	2.368	710	0.018	263.73	2822.6
	Environmental condition Impoverished	-591.24	228.02	-2.592	710	0.009	-1038.9	-143.57
	Boldness of the first baseline	-550.64	649.03	-0.848	710	0.396	-1824.9	723.6
	Bat sex Male	-262	224.84	-1.165	710	0.244	-703.43	179.42
	Bat age	-0.705	1.888	-0.372	710	0.708	-4.414	3.002
	Number of days spent outside until assessment day	10.982	3.868	2.838	710	0.004	3.386	18576
Response	AIC	BIC	Loglikelihood	Deviance				
Explored area	116.55	123.68	-50.277	100.55				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	-4.310	6.312	-0.682	12	0.507	-18.065	9.444
	Environmental condition Impoverished	-5.286	2.081	-2.539	12	0.025	-9.821	-0.750
	Boldness of the first baseline	7.746	6.454	1.200	12	0.253	-6.315	21.809
	Bat sex Male	-2.354	2.260	-1.041	12	0.318	-7.279	2.570
	Bat latest age	0.001	0.022	0.068	12	0.946	-0.047	0.050
	Total amount of nights the bat spent outside	0.183	0.074	2.466	12	0.029	0.021	0.346

Table 4b.

GLMM results for outdoors measurements with the baseline trial PC1 as predisposition.

The lines depicting results for the environmental and the predisposition effects are highlighted in green and blue respectively. response ~ 1 + Environmental_condition + Baseline_PC1 + Sex + Age + Days_spent_outside + (1 | Bat_ID)

Response	AIC	BIC	Loglikelihood	Deviance				
Time spent outside	8636.9	8673.5	-4310.4	8620.9				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	161.22	66.908	2.409	710	0.016	29.864	292.58
	Environmental condition Impoverished	-58.829	26.251	-2.241	710	0.025	-110.37	-7.289
	PC1 of the first baseline	0.619	54.32	0.011	710	0.990	-106.03	107.27
	Bat sex Male	-7.650	26.159	-0.292	710	0.770	-59.01	43.708
	Bat age	0.084	0.242	0.347	710	0.727	-0.391	0.559
	Number of days spent outside until assessment day	1.347	0.379	3.547	710	0.0004	0.601	2.093
Response	AIC	BIC	Loglikelihood	Deviance				
Maximum distance per night	12438	12474	-6210.9	12422				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	1434.7	594.15	2.414	710	0.015	268.2	2601.2
	Environmental condition Impoverished	-580.89	226.37	-2.566	710	0.010	-1025.3	-136.46
	PC1 of the first baseline	-365.17	460.75	-0.792	710	0.428	-1269.8	539.43
	Bat sex Male	-318.9	225.1	-1.416	710	0.157	-760.83	123.03
	Bat age	-1.108	2.121	-0.522	710	0.601	-5.274	3.057
	Number of days spent outside until assessment day	11.783	4.107	2.868	710	0.004	3.718	19.847
Response	AIC	BIC	Loglikelihood	Deviance				
Explored area	117.82	124.94	-50.909	101.82				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	1.331	5.878	0.226	12	0.824	-11.476	14.139
	Environmental condition Impoverished	-5.544	2.154	-2.573	12	0.024	-10.239	-0.849
	PC1 of the first baseline	-1.518	4.323	-0.351	12	0.731	-10.939	7.902
	Bat sex Male	-1.712	2.274	-0.752	12	0.465	-6.668	3.242
	Bat latest age	-0.005	0.025	-0.217	12	0.831	-0.061	0.050
	Total amount of nights the bat spent outside	0.174	0.077	2.243	12	0.044	0.005	0.343

	Pearson r value	Pearson P value
Time spent out vs maximal distance	0.28	0.23
Time spent out vs explored area	0.21	0.38
Maximal distance vs explored area	0.82	<1.269e-05

Table 5.

Pearson correlation test results.

Discussion

Intra-specific inter-individual behavioral differences have been documented in many species^{5,6,8}, but very few studies have explored the processes leading to these differences. In this study, we used a controlled manipulation to examine the effects of the early-life environment to which juvenile fruit-bats had been exposed on their adult behavior.

We demonstrate that early exposure to an enriched environment seems to affect the foraging behavior of bats in the wild. Bats that were exposed early on to an enriched environment were bolder (i.e., flew farther from home), and were more exploratory. Preliminary results suggest that these differences may persist throughout the bat's lifetime; however, additional data must be analyzed to determine the long-term effects. Moreover, their early exposure was a better predictor of their foraging behavior than their individual behavioral predisposition that had been measured indoors before they had any experience of the outside world. We assessed the bats' predispositions at a very young age, as soon as they could fly but before they had a chance to gain any experience. These dispositions are thus likely innate, but we note that they might also have been shaped by maternal effects (e.g., by hormonal transfer, see¹⁰) in addition to genetics. Importantly, all personality traits exhibited significant correlations in the two baseline trials suggesting that indeed they capture behavioral tendencies whose values might change with experience, while individual relations remain similar over time. Moreover, it is possible that a different set of personality traits would have predicted outdoor behavior better than those we used here, although we should note that we chose personality traits commonly used in such studies.

Our findings are in line with the findings from previous research of other organisms that showed that individuals raised in enriched environments^{12,13,15,16} exhibited higher levels of boldness and exploratory behavior. Moreover, studies have shown that exposure to enriched environments enhances motor skills in animals¹³; and growing up in complex environments has been associated with improved learning abilities and with the utilization of multiple cues for obtaining rewards¹⁵. Notably, unlike any previous study, our experimental paradigm allowed us to examine the animals' behavior in their natural habitat following manipulation of their environment early on in life.

The observed behavioral changes in the enriched bats may be the result of increased learning opportunities and enhanced problem-solving abilities driven by early-life exposure to a more dynamic environment. The enriched environment likely provided the bats with more varied experiences, allowing them to practice and refine their foraging and exploratory behaviors. These additional opportunities for learning could have contributed to the increased boldness and exploratory traits observed in the enriched bats. By encountering a more complex environment, these bats may have developed greater behavioral flexibility, enabling them to adapt more efficiently to novel situations in the wild.

Indoor behavior

In contrast to the outdoor behavior, the behavioral traits that we assessed in controlled indoor experiments were not affected by the early environmental enrichment, i.e., enriched bats did not become bolder or more exploratory under laboratory conditions. When examining the bats' behavioral traits across the five trials, we found that boldness and exploration demonstrated a similar temporal dynamics, increasing between consecutive trials that were performed closely to one another in time and decreasing following a prolonged period since the previous trial (**Figure 2**). This pattern indicates that despite the bats' initial habituation to the set-up, following a period of no exposure to the set-up, the bats' boldness (or exploration) again reduced, probably due to some sort of sensitization to the set-up. The bats maintained their within-individual

tendencies while doing so, suggesting that indeed boldness (and to a lesser degree exploration) were good measures of consistent personality. It is possible that a different measure of exploration (one that does not suffer from a ceiling effect like the one we used here) might have shown more consistency over time. Activity, in contrast, continuously (and significantly) showed a different pattern decreasing over time independently of the time elapsed between trials (**Figure 2**). This decrease in activity reflects familiar pups' behavior, which usually becomes less frenetic over time.

The enriched bats were significantly bolder in Trial 5 (**Figure 2**). This could be a real difference indicating an interesting interaction (positive feedback) between the early exposure to an enriched environment with the more exploratory outdoors behavior. That is, the more exploratory tendency of the enriched bats made them even bolder over time. However, we could not overrule the possibility that the difference in trial 5 was a result of a post-selection bias generated by the tendency of bolder individuals in the enriched condition to remain in our colony until the end of the study, while many of the bolder ones from the impoverished condition left the colony earlier and probably moved to other colonies. This tendency can be seen in **Figure 2** when examining only the bats that had remained until the last trial (**Figure 2C1** – and **Figure S2** – see the difference in boldness in trial 2 when only examining the bats that remained in the colony). More research is needed to tell apart these two explanations.

Outdoor tracking

The enriched bats spent significantly more time foraging outdoors, exhibited greater flight distances, and were more exploratory compared to the impoverished bats. These differences could result in actual differences in foraging success. As fruit bats must constantly keep track of an ever-changing landscape of resources (fruit trees)²⁴, being exploratory might pose an advantage for such individuals, especially in a rapidly changing urban environment.

Interestingly, we found no significant correlation between the time bats spent outside and the maximal distance they flew per night. This finding suggests that spending more time outdoors does not necessarily reflect longer foraging journeys; some bats may have remained near the roost despite prolonged activity. This distinction highlights the importance of measuring multiple behavioral dimensions, as time spent outside alone may not accurately capture the spatial scale of foraging behavior.

Research conducted by Egert-Berg et al. (2021), focusing on the foraging behavior of Egyptian fruit bats in the wild, demonstrated that bats that roost in cities display more exploratory behavior than bats that roost in rural habitats. Our current study suggests how such differences in behavior could arise as a result of early-life exposure to an enriched environment such as that in the urban environment, which is much more diverse and dynamic than the rural environment. Urban roosting bats encounter more types of food and must practice more diverse skills to obtain it, like the bats in our enriched colony. Our findings can thus explain how bats and other animals with early exposure to the urban environment become more exploratory. Although fruit-bat pups tend to be more active than adults, it is possible that such adaptations can also be acquired by the adults as well.

The lack of correlation found between the indoor and outdoor behaviors emphasizes the significance of assessing behavior in the bats' natural habitat to obtain more reliable and meaningful results. While measuring personality traits, particularly boldness and exploration, in a small captive setup clearly assesses some aspects of personality, translating these to behavior in the real world may be limited. It is possible that other behavioral traits measured indoors, such as neophobia or decision-making under uncertainty, might have shown stronger associations with outdoor behavior. Although some studies have found an indoor/outdoor correlation in other

animals (see [25](#)), our findings highlight the importance of choosing behavioral traits that are ecologically relevant, and suggest that the traits we measured may not have captured the dimensions most predictive of natural behavior.

Overall, our study provides evidence of the profound impact of early life environmental conditions on the behavior of the adult bats. These findings contribute to our understanding of the developmental factors that shape animal behavior and highlight the importance of environmental enrichment during the early life stage.

Methods

Ethical statement

Experiments were approved by the TAU IACUC; permit number: 04-18-030. Capture of Egyptian fruit bats for the research was approved by the Israel National Park Authority, permit number 2020/42646.

Experimental animals

The experiment was conducted on young Egyptian fruit bats. A total of 40 bats were captured from three wild colonies (Herzliya, Beit Guvrin, and Tinsmet; [Table 6](#)) between the years 2019-2021 and housed in the Zoological Garden at Tel Aviv University. We captured pregnant females or females with young pups before their volant stage, to ensure that the pups would be naïve and without previous individual flight experience. Of the 40 bats in the study, 21 were born in our lab and 19 were captured at a very young age together with their mothers, before they could fly independently. To avoid an origin bias, bats from all colonies were divided equally and housed in two identical rooms ($\sim 2.5 \times 2.5 \text{ m}^3$, [Table 6](#)), maintained at a controlled temperature of 24-26°C and a light cycle approximately matching the times of sunrise/sunset. The bats' diet in both environments comprised seasonal mixed fruits (watermelon, apples, melons, and bananas *ad lib*).

Upon arrival, the bats' weight and forearm were measured and documented. Each mother-pup pair was bleached with a unique mark, and each bat was also tagged with a subcutaneous electronic RFID chip. All bats were measured and checked regularly every 2-3 weeks to make sure they were healthy. Three individuals were removed from the experiment: two due to injury (broken wings), and one due to weight loss (>10% loss of their previous body weight).

Environmental conditions

While both colonies were provided with the same basic conditions and food, the bats in the “enriched” environment regularly experienced varying environmental enrichment three times a week, including changing the number and shape of food bowls, hanging food baskets from the ceiling, playback of vocalizations recorded from a larger colony, and adding rope ladders or ropes hanging from the ceiling. In the “impoverished” environment the animals experienced a dull environment with no variation in conditions throughout the study period. They received their food in one bowl, while nothing was hung from the ceiling and no other enrichment was provided.

Defining baseline personality in juveniles – the foraging box experiment

As juveniles (55-281 days old), the young bats participated in a foraging box experiment to examine a set of personality traits in a captive setting [10](#). The experimental set-up consisted of an indoor tent ($3.9 \times 2.7 \times 1.9 \text{ m}^3$), with six evenly spaced black plastic boxes placed on the floor (i.e. foraging box, $64 \times 38 \times 40 \text{ cm}^3$). Each box had a round hole in its lid (10 cm diameter) with a mesh

Season	Environmental condition	Colony ID	Coordinates	Number of pairs
1 (2019-2020)	Enriched	Tinshemet	31°59'43.2"N 34°57'19.2"E	1
		Beit Guvrin	31°36'45.8"N 34°53'41.7"E	1
		Herzliya	32°10'18.4"N 34°48'51.1"E	4
	Impoverished	Tinshemet	31°59'43.2"N 34°57'19.2"E	1
		Beit Guvrin	31°36'45.8"N 34°53'41.7"E	1
		Herzliya	32°10'18.4"N 34°48'51.1"E	3
2 (2020-2021)	Enriched	Beit Guvrin	31°36'45.8"N 34°53'41.7"E	5
		Herzliya	32°10'18.4"N 34°48'51.1"E	9
	Impoverished	Beit Guvrin	31°36'45.8"N 34°53'41.7"E	5
		Herzliya	32°10'18.4"N 34°48'51.1"E	10

Table 6.

Egyptian Fruit bat pups were captured with their mothers in rural and urban colonies and then divided between two environmental conditions, enriched and impoverished.

ladder leading down to a food bowl, containing sufficient amount of food for the bat for an entire night, to prevent the bat from selecting another box simply because the first one is empty (**figure 4**).

The exposure phase. Bats are social animals ^{26,27}. To reduce the stress of being alone in a new environment before the first trial, a group of 4-5 pups from the same environment was placed in the tent overnight (16:00-8:00) for a session of exposure. Each of the six boxes contained an amount of food sufficient for all of the bats taking part in the exposure.

The Experiment. Following this one night of introduction, each pup (n=39, see **Table 8**) was placed alone in an open carrying bag (35x26x30 cm³) inside the flight tent with the six foraging boxes (described above) for a full night. Each food bowl offered an amount of food that was enough for the full night, containing 25 pieces of fruit (with a total weight of 150 g) and 50 ml mango nectar. The experiment took place two nights in a row to determine behavioral consistency. The experiment was recorded using an IR video camera (Sony HDRCX730, Sony FDR-AX53), together with one infrared light (Methaphase Technologies Inc-ISO-14-IR-24) placed inside the tent.

The environmental enrichment began only after all the bats in the enriched room had completed the baseline experiments (see **Table 7** for the full time-line of the experiment). Both groups then experienced the (enriched/impoverished) environment for at least two months before moving on to the next stage the open colony, from which the bats are free to fly in and out as they choose (see details below).

Personality after foraging in the wild

Bats that routinely exited the open colony to forage and returned were twice tested again in the foraging box experiment: 2-3 weeks following their first exit (Trial 4, post-release 1), and 5-6 weeks following their first exit (Trial 5, post-release 2). A total of 19 bats were tested for these trials (see **Table 8**).

Behavioral analysis

Each bat's actions during the foraging box experiment were documented based on video recordings by two independent observers. In cases of disagreement, a third observer compared the two and made the final judgement. The identified actions comprised: landing on or entering the carrying bag in which the bats had been introduced into the room; landing; entering the food boxes; collecting food; or flying around to explore the environment.

Similar to our previous study ¹⁰, we then calculated the following personality traits:

- **Boldness/Risk-taking:** boldness can be defined in several ways ²⁸. We measured it as the proportion of times the bat entered the boxes out of the total number of landings and entering with or without collecting food from the boxes. A previous study has shown that entering boxes placed on the ground is a risky behavior that a bat generally avoids ¹⁰.
- **Exploration:** the exploratory index was calculated as the number of individual foraging boxes the bat visited (i.e., 0-6). Note that each box contained enough food for the entire night eliminating the need for it to visit more than one box.
- **Activity:** activity level was defined as the number of interactions with the boxes (landings and entries) normalized by the total time in the experiment (i.e., after the bat had left the carrying bag for the first time).

Figure 4.

Experimental set-up schematic.

Each bat is placed in a tent with 6 foraging boxes

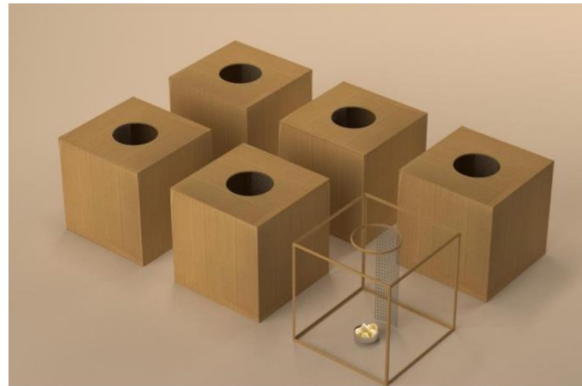


Table 7.

Full time-line of the experiment.

	October-December				December-June	
2019-2020	Baseline Trials N=11	Environmental enrichment		Release to open colony	Tracking bats using GPS devices N=10	Post release trials N=9
2020-2021	Baseline Trials N=29	Environmental enrichment	Post-enrichment trial N=29	Release to open colony	Tracking bats using GPS devices N=11	Post release trials N=10

During the second season (year 2020-21), following the exposure to the environmental conditions, all bats (n=29, Table 8) underwent an additional night in the foraging box experiment to evaluate the effect of the enriched environment. This additional trial took place prior to their transfer to the open colony and is termed the “post-enrichment trial”.

	Environmental condition	Baseline (both 1+2)	Post-enrichment trial	Post release trials (4+5)
Season 1	Enriched	6	-	4
	Impoverished	5	-	5
Season 2	Enriched	13	14	5
	Impoverished	15	15	5
Total number of bats		39	29	19

Table 8.

Number of bats that participated in each of the trials.

Statistical analysis of the behavioral trials

To verify that there were no behavioral differences between the two groups prior to applying our environmental conditions, we compared the baseline trials between the two environmental conditions (**Table S1** [1](#)). We used mixed-effect general linear model (GLMM) implemented in MATLAB R2019, with the traits set as the response variable, the environmental condition and trial number (baseline 1 or 2) set as fixed factors, and bat ID set as random effect (n=39, **Table 8** [2](#)).

We use the term "personality" to encompass a set of behavioral traits that exhibit consistency within individuals over time and across various contexts [1](#). To evaluate the consistency of the measured behavioral traits, we conducted a comparison between the bats' first trial (baseline 1) and the trial conducted following their exposure to the different environmental conditions (post-enrichment trial). These analyses incorporated only the individuals from the second measuring season, because we only ran the post-enrichment measurement during the second season.

To assess the impact of the environmental condition on each behavioral trait, we employed GLMMs. We used the behavioral trait as the response parameter, the tested environmental condition (i.e., enriched or impoverished), trial number (1-5), and the interaction between environmental conditions and trial number as fixed factors. Bat ID was set as a random effect.

Analyzing the bats' outdoor foraging behavior

Following the captive stage described above, bats were transferred to our open colony, an Egyptian fruit bat colony, representing a natural urban roost where bats can exit nightly to forage in the wild and return to roost in our fully monitored day-roost (or move to a different roost). Food is provided in the roost every evening. This system allows us to continuously monitor the bats' natural behaviors within and outside the roost [29](#).

The bats were released in groups of 4-6 individuals (at an average age of 218.92 ± 56.8 days) from the same experimental colony (either the enriched or impoverished environment), alternately between these two environmental conditions. Each bat was fitted with an on-board GPS device (Vesper, ASD inc.) following its 2nd foraging night in the wild, in order to monitor most of the navigational history of all individuals. The device was coated using Parafilm (Heathrow Scientific) and duct tape, and was attached to the bats using a chain necklace covered with heat-shrink (see full details in [30](#)). The mean weight mounted on the bats was 6 gr, accounting for $5.6\% \pm 0.6\%$ of the bats' weight. The GPS was programmed to start recording a few hours before sunset and stopped at 6:00AM, collecting GPS locations every 30 seconds.

These data allowed us to assess various parameters extracted from the individuals' natural foraging behavior, which served as proxies for their exploratory and activity tendencies. Specifically, we measured the:

- **Maximum flying distance:** Defined as the distance to the furthest point from the open colony that the bat reached on every night outside.
- **Time spent outdoors:** Defined as the total time the bat spent outside the colony on a given night. Nights when a bat did not leave the colony (6.2 days per bat on average) were not included in the analysis.
- **The explored area** was estimated as follows: For each bat, we calculated the maximum distance from the colony for each night. We then computed the 95th percentile of these nightly maximum distances and used it as a threshold radius. Using the colony as the center point, we excluded all GPS locations falling beyond this radius. A convex hull was then calculated from the remaining points to estimate the core area utilized by each bat.

Statistics: To examine the influence of the environmental condition (i.e., enriched or impoverished), we used GLMs for each of the three outdoor traits, with the traits set as the response parameter. Environmental condition, bat age and sex, and the number of nights the bat spent outside until the assessment day were set as a fixed effect. For the explored area analysis, the total number of days the bat spent outside during the entire tracking period and the bat's age at the last time of measuring were set as fixed effects. For the maximum distance and the time spent outdoors, we used data from each night the bat was outside. Explored area receiving only one (seasonal) measurement for each bat.

Captivity and outdoor comparison

We sought to demonstrate the effectiveness of our captive setup by comparing the individual's behavior in both indoor and outdoor settings. To achieve this, we analyzed the Pearson correlation between the first post-release trial and the average of the subsequent five days of outdoor measurements using Pearson correlation test. In addition, we tested the correlation between the individual's first post-release trial and the average of its subsequent 20 days outdoor measurements.

Availability of data and materials

The datasets generated and analyzed during the current study are available on Mendeley Data: DOI: 10.17632/wh7c636y3t.1

Acknowledgements

We would like to thank Mor Taub for assistance with figure design, Goni Naamani for assisting with the data analysis. This project was partially funded by the ERC project BehaviorIsland.

Additional information

Author contribution

A.R., Y.Y., and L.H. Conceived and designed the research. A.R, R.A and N.G. collected the data. A.R., A.G., and X.C. analyzed the data. A.R. and Y.Y. wrote the manuscript. A.G. reviewed and edited the manuscript. All authors read and approved the final manuscript.

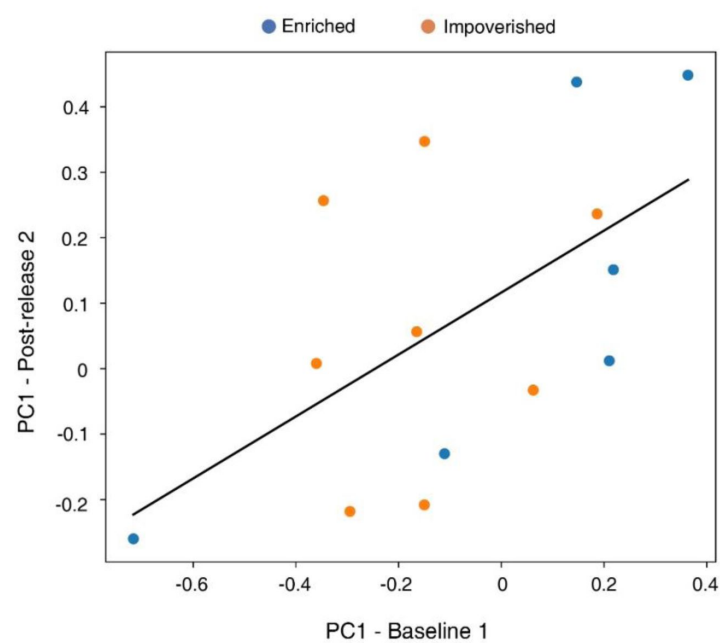


Figure S1.

PCA analysis for trial 1 (baseline) and 5 (post-release 2); positive correlation was found between the individual scores on PC1 (Pearson correlation $P=0.03$, $r=0.58$).

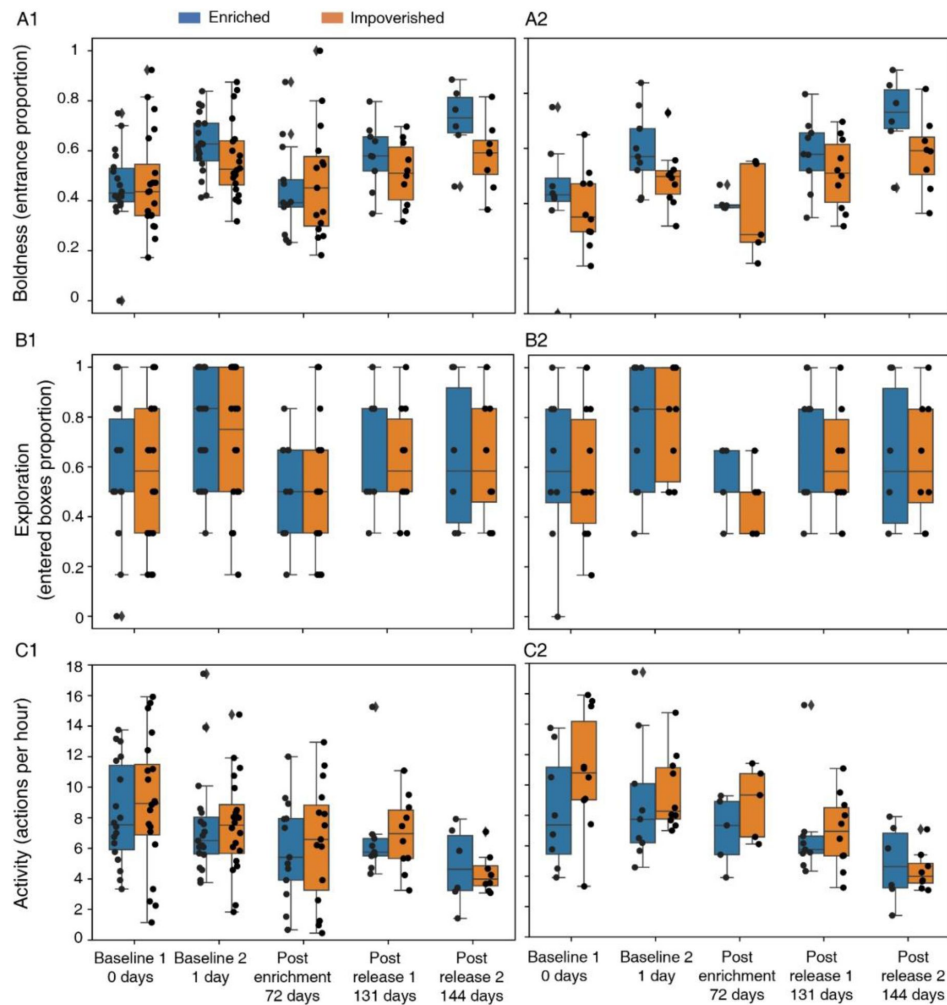


Figure S2.

Full bat behavioral traits over time.

Box plots depicting the bats' behavior across the five trials according to three traits: Boldness (A), exploration (B), and activity levels (C). The experimental phase and the time in days from the day of the first trial are depicted on the X-axis. Box plots show the 25% and 75% percentiles. Medians and whiskers based on 1.5 IQR are shown. A1, B1, and C1 represent all bats tested, while A2, B2, and C2 represent only the bats that were tracked outdoors using GPS devices.

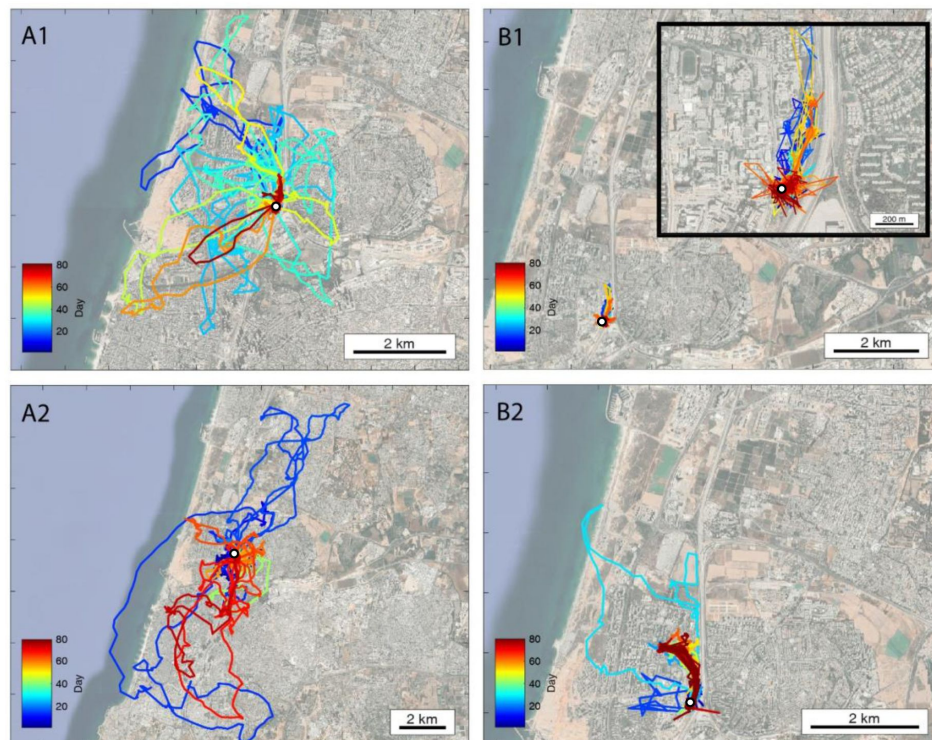


Figure S3.

Early environmental exposure affects outdoor foraging behavior.

The complete movement of four individuals - two from each colony. **(A)** Individuals raised in the enriched colony, data shown for two individuals (A1,A2); **(B)** Individuals raised in the impoverished colony, data shown for two individuals (B1,B2). Colors depict time in days. The open colony is marked by a white circle. Insert in B1 zooms-in on the tracks.

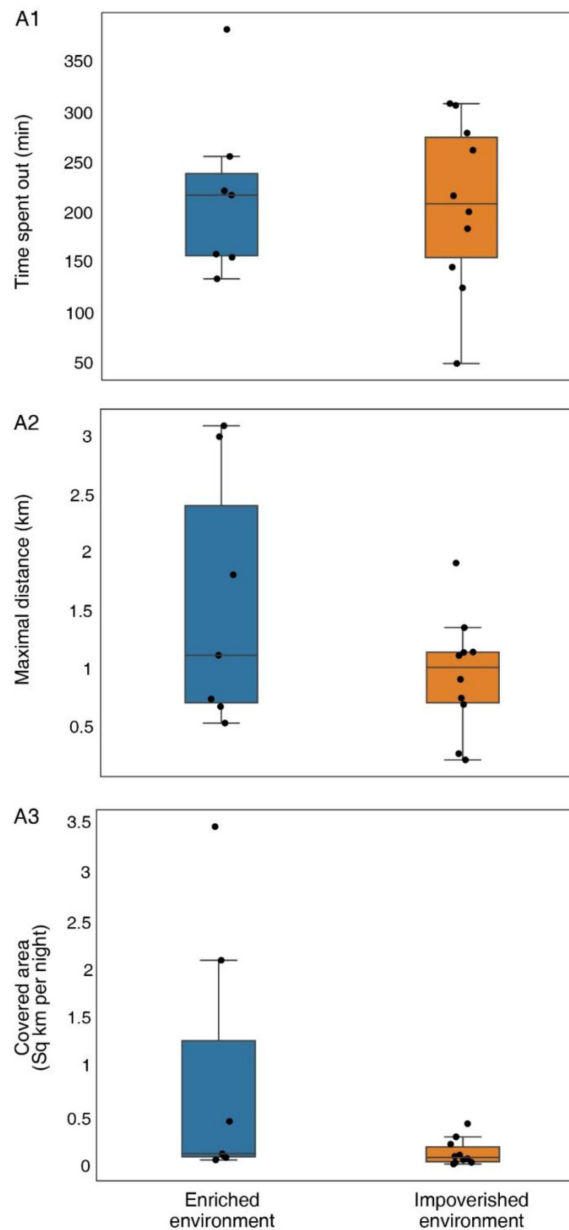


Figure S4.

Early environmental exposure affects outdoor foraging behavior.

(A1) Time spent by the bats outside the colony each night (in minutes). **(A2)** Distance to the furthest point from the colony per night. **(A3)** The area explored by the bats. The data for each bat were estimated for the period between its 21st and 26th day outdoors (n=17 bats). Median and whiskers based on 1.5 IQR are shown.

Response	AIC	BIC	LogLikelihood	Deviance				
Boldness (N=40)	-74.127	-62.28	42.064	-84.127				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.340	0.047	7.244	76	3.031e-10	0.246	0.434
	Environmental condition	-0.022	0.044	-0.497	76	0.620	-0.110	0.066
	Trial number (1 or 2)	0.133	0.023	5.776	76	1.581e-07	0.087	0.178
Response	AIC	BIC	LogLikelihood	Deviance				
Boldness (N=19)	-30.614	-22.559	20.307	-40.614				
	Fixed effects coefficients (95% CIs)							

Table S1.

**Mixed GLM results for baseline trials of two environmental conditions
response ~ 1 + Environmental_condition + Trial_number + (1| Bat_ID)'.**

	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.303	0.071	4.239	34	0.0001	0.157	0.448
	Environmental condition Impoverished	-0.071	0.055	-1.303	34	0.201	-0.184	0.04
	Trial number (1 or 2)	0.137	0.038	3.59	34	0.001	0.059	0.215
Response	AIC	BIC	LogLikelihood	Deviance				
	-8.551	3.295	9.275	-18.551				
Exploration (N=40)	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.416	0.071	5.854	76	1.145e-07	0.274	0.558
	Environmental condition	-0.014	0.074	-0.199	76	0.842	-0.163	0.133
	Trial number (1 or 2)	0.160	0.031	5.111	76	2.312e-06	0.097	0.222
Response	AIC	BIC	LogLikelihood	Deviance				
	6.103	14.158	1.948	-3.896				
Exploration (N=19)	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.392	0.113	3.444	34	0.001	0.161	0.623
	Environmental condition Impoverished	-0.003	0.103	-0.033	34	0.973	-0.213	0.206
	Trial number (1 or 2)	0.190	0.055	3.463	34	0.001	0.078	0.302
Response	AIC	BIC	LogLikelihood	Deviance				
	-874.680	-862.840	442.360	-884.680				
Activity level (N=40)	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.002	0.0002	8.993	76	1.371e-13	0.002	0.003
	Environmental condition	0.000	0.0002	0.358	76	0.721	-0.0004	0.0006
	Trial number (1 or 2)	-0.000	0.0001	-2.175	76	0.032	-0.0006	-2.78e-05
Response	AIC	BIC	LogLikelihood	Deviance				
	-399.320	-391.270	204.660	-409.320				
Activity level (N=19)	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	0.002	0.0005	5.118	34	1.205e-05	0.001	0.003
	Environmental condition	0.000	0.0003	1.186	34	0.243	-0.0002	0.001
	Trial number (1 or 2)	-0.000	0.0002	-0.419	34	0.677	-0.0006	0.0004

Table S1. (continued)

Response	AIC	BIC	LogLikelihood	Deviance				
Proportion of Nights Outside	-207.95	-185.13	108.97	-217.95				
	Fixed effects coefficients (95% CIs)							
	Name	Estimate	SE	tStat	DF	P value	Lower	Upper
	Intercept	1.315	0.419	3.134	705	0.001	0.491	2.139
	Environmental condition_Impoverished	0.024	0.399	0.061	705	0.951	-0.760	0.808
	Sex_M	0.323	0.393	0.820	705	0.412	-0.450	1.096
	Age	6.77e-05	0.0003	0.179	705	0.857	-0.0006	0.0008

Table S2.

GLMM results for proportion of nights foraging outside the roost (%).

DaysOutProportion ~ 1 + EnvironmentalCondition + Sex + Age + (1 | Bat_ID) With Y ~ Binomial (TotalExperimentDays) and link=logit

References

1. Stamps J., Groothuis T. G. G (2010) **The development of animal personality: relevance, concepts and perspectives** *Biol. Rev* **85**:301–325 [Google Scholar](#)
2. Beekman M., Jordan L. A (2017) **Does the field of animal personality provide any new insights for behavioral ecology?** *Behav. Ecol* **28**:617–623 [Google Scholar](#)
3. Hertel A. G., Niemelä P. T., Dingemanse N. J., Mueller T (2020) **A guide for studying among-individual behavioral variation from movement data in the wild** *Mov Ecol* **8** [Google Scholar](#)
4. Roche D. G., Careau V., Binning S. A (2016) **Demystifying animal ‘personality’ (or not): why individual variation matters to experimental biologists** *J. Exp. Biol.* **146**:712 <https://doi.org/10.1242/jeb.146712> | [Google Scholar](#)
5. Kazlauskas V., et al. (2005) **Behavioral and cognitive profile of mice with high and low exploratory phenotypes** *Behav. Brain Res* **162**:272–278 [Google Scholar](#)
6. Menzies A. K., Timonin M. E., McGuire L. P., Willis C. K. R (2013) **Personality Variation in Little Brown Bats** *PLoS ONE* **8**:e80230 [Google Scholar](#)
7. Réale D., Reader S. M., Sol D., McDougall P. T., Dingemanse N. J (2007) **Integrating animal temperament within ecology and evolution** *Biol. Rev* **82**:291–318 [Google Scholar](#)
8. Freeman H. D., Gosling S. D (2010) **Personality in nonhuman primates: a review and evaluation of past research** *Am. J. Primatol* **72**:653–671 [Google Scholar](#)
9. Speechley E. M., et al. (2024) **Aggressive interactions influence cognitive performance in Western Australian magpies** *Proc. R. Soc. B Biol. Sci* **291** [Google Scholar](#)
10. Harten L., Gonceer N., Handel M., Dash O., Fokidis H. B., Yovel Y (2021) **Urban bat pups take after their mothers and are bolder and faster learners than rural pups** *Bmc biology* **19**:1–10 [Google Scholar](#)
11. Guenther A., Finkemeier M.-A., Trillmich F (2014) **The ontogeny of personality in the wild guinea pig** *Anim Behav* **90**:131–139 [Google Scholar](#)
12. Xu W., et al. (2021) **Environmental complexity during early life shapes average behavior in adulthood** *Behav. Ecol* **32**:105–113 [Google Scholar](#)
13. Crawford L. E., et al. (2020) **Enriched environment exposure accelerates rodent driving skills** *Behav. Brain Res* **378** [Google Scholar](#)
14. Díaz-Fleischer F., Arredondo J., Aluja M (2009) **Enriching early adult environment affects the copulation behaviour of a tephritid fly** *J. Exp. Biol* **212**:2120–2127 [Google Scholar](#)
15. White G. E., Brown C (2015) **Cue choice and spatial learning ability are affected by habitat complexity in intertidal gobies** *Behav. Ecol* **26**:178–184 [Google Scholar](#)

16. Zaias J., Queeney T. J., Kelley J. B., Zakharova E. S., Izenwasser S (2008) **Social and Physical Environmental Enrichment Differentially Affect Growth and Activity of Preadolescent and Adolescent Male Rats** *J. Am. Assoc. Lab. Anim. Sci* **47** [Google Scholar](#)
17. Herborn K. A., et al. (2010) **Personality in captivity reflects personality in the wild** *Anim Behav* **79**:835–843 [Google Scholar](#)
18. Harten L., et al. (2018) **Persistent producer-scrourer relationships in bats** *Sci. Adv* **4**:e1603293 [Google Scholar](#)
19. Harten L., Katz A., Goldshtein A., Handel M., Yovel Y (2020) **The ontogeny of a mammalian cognitive map in the real world** *Science* **369**:194–197 [Google Scholar](#)
20. Egert-Berg K., et al. (2021) **Fruit bats adjust their foraging strategies to urban environments to diversify their diet** *BMC Biol* **19** [Google Scholar](#)
21. Fehlmann G., et al. (2017) **Extreme behavioural shifts by baboons exploiting risky, resource-rich, human-modified environments** *Sci. Rep* **7** [Google Scholar](#)
22. Weimerskirch H., Corbeau A., Pajot A., Patrick S. C., Collet J (2023) **Albatrosses develop attraction to fishing vessels during immaturity but avoid them at old age** *Proc. R. Soc. B Biol. Sci* **290** [Google Scholar](#)
23. Shoval O., et al. (2012) **Evolutionary Trade-Offs, Pareto Optimality, and the Geometry of Phenotype Space** *Science* **336**:1157–1160 [Google Scholar](#)
24. Harten L., et al. (2024) **Time-mapping and future-oriented behavior in free-ranging wild fruit bats** *Curr. Biol* <https://doi.org/10.1016/j.cub.2024.05.046> | [Google Scholar](#)
25. Herborn K. A., et al. (2010) **Personality in captivity reflects personality in the wild** *Anim Behav* **79**:835–843 [Google Scholar](#)
26. Kerth G (2008) **Causes and Consequences of Sociality in Bats** *BioScience* **58**:737–746 [Google Scholar](#)
27. Kwiecinski G. G., Griffiths T. A. (1999) **Rousettus egyptiacus**. *Mamm Species* **1** <https://doi.org/10.2307/3504411> | [Google Scholar](#)
28. Carter A. J., Feeney W. E., Marshall H. H., Cowlshaw G., Heinsohn R (2013) **Animal personality: what are behavioural ecologists measuring?: What are animal personality researchers measuring** *Biol. Rev* **88**:465–475 [Google Scholar](#)
29. Harten L., Katz A., Goldshtein A., Handel M., Yovel Y (2020) **The ontogeny of a mammalian cognitive map in the real world** *Science* **369**:194–197 [Google Scholar](#)
30. Goldshtein A., Harten L., Yovel Y (2022) **Mother bats facilitate pup navigation learning** *Curr. Biol* **32**:350–360 [Google Scholar](#)

Author information

Adi Rachum

School of Zoology, Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel
ORCID iD: [0009-0007-2494-0602](https://orcid.org/0009-0007-2494-0602)

Lee Harten

School of Zoology, Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel

Reut Assa

School of Zoology, Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel

Aya Goldshtein[&]

School of Zoology, Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel

[&]Present address: Max Planck Institute of Animal Behavior, Department of Collective Behavior, Konstanz, Germany

Xing Chen

School of Zoology, Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel

Nesim Gonceer

School of Zoology, Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel

Yossi Yovel

School of Zoology, Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel, Sagol School of Neuroscience, Tel Aviv University, Tel Aviv, Israel
ORCID iD: [0000-0001-5429-9245](https://orcid.org/0000-0001-5429-9245)

For correspondence: yossiyovel@gmail.com

Editors

Reviewing Editor

Kate Wassum

University of California, Los Angeles, Los Angeles, United States of America

Senior Editor

Kate Wassum

University of California, Los Angeles, Los Angeles, United States of America

Reviewer #1 (Public review):

Summary:

The authors show that early life experience of juvenile bats shape their outdoor foraging behaviors. They achieve this by raising juvenile bats either in an impoverished or enriched environment. They subsequently test the behavior of bats indoors and outdoors. The authors show that behavioral measures outdoors were more reliable in delineating the effect of early life experiences as the bats raised in enriched environments were more bold, active and exhibit higher exploratory tendencies.

Strengths:

The major strength of the study is providing a quantitative study of animal "personality" and how it is likely shaped by innate and environmental conditions. The other major strength is the ability to do reliable long term recording of bats in the outdoors giving researchers the opportunity to study bats in their natural habitat. To this point, the study also shows that the behavioral variables measured indoors do not correlate to that measured outdoor, thus providing a key insight into the importance of test animal behaviors in their natural habitat.

Weaknesses were in the first round of review:

It is not clear from the analysis presented in the paper how persistent those environmentally induced changes, do they remain with the bats till end of their lives.

Comments on revisions:

The authors have addressed those weaknesses and the paper is much stronger.

<https://doi.org/10.7554/eLife.103220.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Reviewer):

It is not clear from the analysis presented in the paper how persistent those environmentally induced changes, do they remain with the bats till the end of their lives.

Currently, the long-term effects of enrichment on the bats remain uncertain. Preliminary results suggest that these differences may persist throughout the bats' lifetimes; however, further data analysis is ongoing to determine the extent of these effects. We also addressed now at the manuscript discussion

Reviewer #2 (Public Reviewer):

(1) Assessing personality metrics and the indoor paradigm: While I applaud this effort and think the metrics used are justified, I see a few issues in the results as they are currently presented:

(a) [Major] I am somewhat concerned that here, the foraging box paradigm is being used for two somewhat conflicting purposes: (1) assessing innate personality and (2) measuring changes in personality as a result of experience. If the indoor foraging task is indeed meant to measure and reflect both at the same time, then perhaps this can be made more explicit throughout the manuscript. In this circumstance, I think the authors could place more emphasis on the fact that the task, at later trials/measurements, begins to take on the character of a "composite" measure of personality and experience.

Personality traits should generally be stable over time, but personality can also somewhat change with experience. We used the foraging box to assess individual personality, but we also examined the assumption that what we are measuring is a proxy of personality and hence is stable over time. We now clarify this in the manuscript.

(b) [Major] Although you only refer to results obtained in trials 1 and 2 when trying to estimate "innate personality" effects, I am a little worried that the paradigm used to measure personality, i.e. the stable components of behavior, is itself affected by other

factors such as age (in the case of activity, Fig. 1C3, S1C1-2), the environment (see data re trial 3), and experience outdoors (see data re trials 4/5).

We found that boldness was the most consistent trait, showing persistence between trials 1 to 5, i.e., 144 days apart on average. We thus also used Boldness as the primary parameter for assessing the effects of personality on the outdoors behavior. While we evaluated other traits for completeness, boldness was the only one that consistently met the criteria for personality, which is why we focused on it in our analyses. The other traits which were not stable over time could be used to assess the effects of experience on behavior

Ideally, a study that aims to disentangle the role of predisposition from early-life experience would have a metric for predisposition that is relatively unchanging for individuals, which can stand as a baseline against a separate metric that reflects behavioral differences accumulated as a result of experience.

I would find it more convincing that the foraging box paradigm can be used to measure personality if it could be shown that young bats' behavior was consistent across retests in the box paradigm prior to any environmental exposure across many baseline trials (i.e. more than 2), and that these "initial settings" were constant for individuals. I think it would be important to show that personality is consistent across baseline trials 1 and 2. This could be done, for example, by reproducing the plots in Fig. 1C1-3 while plotting trial 1 against trial 2. (I would note here that if a significant, positive correlation were to be found (as I would expect) between the measures across trial 1 and 2, it is likely that we would see the "habituation effect" the authors refer to expressed as a steep positive slope on the correlation line (indicating that bold individuals on trial 1 are much bolder on trial 2).)

We agree and thus used boldness which was found to be stable over five trials (three of which were without external experience). We note that if Boldness as we measured it increased over time, the differences between individuals remained similar and this is what is expected from personality traits measured in the same paradigm several times (after the animal acquires experience).

(c) Related to the previous point, it was not clear to me why the data from trial 2 (the second baseline trial) was not presented in the main body of the paper, and only data from trial 1 was used as a baseline.

We added a main figure, showing the correlation between the two baseline trials

In the supplementary figure and table, you show that the bats tended to exhibit more boldness and exploratory behavior, but fewer actions, in trial 2 as compared with trial 1. You explain that this may be due to habituation to the experimental setup, however, the precise motivation for excluding data from trial 2 from the primary analyses is not stated. I would strongly encourage the authors to include a comparison of the data between the baseline trials in their primary analysis (see above), combine the information from these trials to form a composite baseline against which further analyses are performed, or further justify the exclusion of data as a baseline.

We had no intention of excluding data from baseline 2. As we have shown several times before (e.g., Harten, 2021) bats' boldness as we measure it in the box experiment increases over sessions performed nearby in time. This means that trial 2's boldness was higher than that of trial 1 and trial 3 which made the data less suitable for a Linear model. Moreover, our measurement of boldness is capped (with a maximum of 1) again making it less suitable for a Linear model. However, following the reviewer's question we now ran all analyses with trial

2's data included and not only that the results remained the same, some of the models fit better (based on the AIC criterion). We added this information to the revised manuscript.

(2) Comparison of indoor behavioral measures and outdoor behavioral measures
Regarding the final point in the results, correlation between indoor personality on Trial 4 and outdoor foraging behavior: It is not entirely clear to me what is being tested (neither the details of the tests nor the data or a figure are plotted). Given some of the strong trends in the data - namely, (1) how strongly early environment seems to affect outdoor behavior, (2) how strongly outdoor experience affects boldness, measured on indoor behavior (Fig. 1D) - I am not convinced that there is no relationship, as is stated here, between indoor and outdoor behavior. If this conclusion is made purely on the basis of a p-value, I would suggest revisiting this analysis.

We agree that the relationship between indoor personality measures and outdoor foraging behavior is of great interest and had expected to find some correspondence between the two. To test this, we conducted multiple GLM analyses using the different indoor behavioral traits as predictors of outdoor behaviors. These analyses did not reveal any significant correlations. We also performed a separate analysis using PC1 (derived from the indoor behavioral variables) as a predictor, and again found no significant associations with outdoor behavior.

We were indeed surprised by this outcome. It is possible that the behavioral traits we assessed indoors (boldness, exploration, and activity) do not fully capture the dimensions of behavior that are most relevant to foraging in the wild. For example, traits such as neophobia or decisionmaking under risk, which we did not assess directly, may have had stronger predictive value for outdoor behavior. We now highlight this point more clearly in the Discussion and acknowledge the possibility that alternative or additional personality traits might have revealed meaningful relationships.

(3) Use of statistics/points regarding the generalized linear models While I think the implementation of the GLMM models is correct, I am not certain that the interpretation of the GLMM results is entirely correct for cases where multivariate regression has been performed (Tables 4s and S1, and possibly Table 3). (You do not present the exact equation they used for each model (this would be a helpful addition to the methods), therefore it is somewhat difficult to evaluate if the following critique properly applies, however...)

The "estimate" for a fixed effect in a regression table gives the difference in the outcome variable for a 1 unit increase in the predictor variable (in the case of numeric predictors) or for each successive "level" or treatment (in the case of categorical variables), compared to the baseline, the intercept, which reflects the value of the outcome variable given by the combination of the first value/level of all predictors. Therefore, for example, in Table 4a - Time spend outside: the estimate for Bat sex: male indicates (I believe) the difference in time spent outside for an enriched male vs. an enriched female, not, as the authors seem to aim to explain, the effect of sex overall. Note that the interpretation of the first entry, Environmental condition: impoverished, is correct. I refer the authors to the section "Multiple treatments and interactions" on p. 11 of this guide to evaluating contrasts in G/LMMS: <https://bbolker.github.io/mixedmodelsmisc/notes/contrasts.pdf>

We are not certain we fully understand the comment; however, if our understanding is correct, we respectfully disagree. A GLM analysis without interaction terms—as conducted in our study—functions as a multiple linear regression, wherein each factor's estimate reflects its individual effect on the dependent variable. For example in the case of sex, it examines the effect of sex on the time spent out independently of enrichment. An interaction term would be needed to test sex*enrichment. We have added the models' formula, and we hope this clarifies our approach

Reviewer #1 (Recommendations for the authors):

I would recommend the following:

(1) As video tracking and behavioral analysis softwares are wide spread, it would be great to see this applied to the bat behavior indoor to answer questions like how does the bat velocity or heading or acceleration correlate with the behavioral measures boldness , activity or exploration? In the same gist, can one infer boldness, activity or exploration from measured bat velocity or other parameters? I think this will further make the indoor behavior more quantitative.

In a tent of the size used in our study, bats' flight behavior tends to be highly stereotypical: they typically perch on the wall, take off, circle the tent—sometimes multiple times—and then either land or not, and enter or not. Flight velocity is largely determined by individual maneuverability and the physical constraints of the space; thus, precise tracking is unlikely to provide further insight into boldness. In contrast, decision-making behaviors—such as whether to land or enter—more accurately reflect personality traits, as we have shown previously (Harten et al., 2018). Moreover, accurate 3D tracking in such an environment is possible but definitely not easy due to the many blind-spots resulting from the cameras being inside the 3D volume. Nonetheless, we quantified flight activity and assessed its correlation with the other behavioral axes. As it was highly correlated with general activity, we did not include it as an independent parameter in the main analysis. However, in response to the reviewer's suggestion, we now present this analysis in the Supplementary Materials.

(2) It is not clear whether the bats come from the same genetic background. they might be but it is not mentioned in the methods under the experimental subjects.

We have shown in the past that there is no familial relations in a randomly caught sample of bats in the colony where we usually work (Harten et al., 2018). The bats were caught in three, not related wild colonies. The text referring to the table was clarified in the revised manuscript

(3) It will be great to include the author's thoughts about mechanisms underlying those environmentally induced changes in behavior in the discussion section along with how this will affect the bats' social foraging abilities. Another question that comes to mind is whether growing up with a large number of bats constitute an enriched environment in itself.

We agree that this could count as an enrichment, and we thus ensured similar group sizes in both groups for this reason. We clarify this in the revised manuscript.

We have elaborated on the underlying mechanisms in the discussion, focusing on how they contribute to behavioral changes.

Reviewer #2 (Recommendations for the authors):

(1) Outdoor foraging behavior

If I understand correctly, the data you display in Fig. 3A is only from the 2nd to 3rd weeks of exploration, i.e. just before the first post-exploration trial.

What does the data look like for the second outdoor exploration data, i.e. before the final trial?

Is there a specific reason why these measures were only computed on the GPS data from the 3rd week outside? If so, can this sampling of the data be motivated or briefly addressed (in the methods and wherever else necessary)?

In order to allow a comparison between individuals, we had to restrict ourself to a period we had data from many individuals (some disappeared later on).

Following the reviewer suggestion – we added a supplementary figure including days 21-26

I would find it important and of great interest to see movement maps for more animals, as these give very rich information that is not entirely captured by the three proxies of outdoor activity.

Are these four exemplary animals sampled from both seasons?

Did you check to see if there were any overall differences in outdoor foraging behavior as a function of the season in which the bats were captured?

Yes, the samples represent individuals from both tested years. This was clarified, and additional examples were included in a supplementary figure.

Variable of time spent outdoors: You mention that you did not include the nights that the bat spent in the colony in these calculations. Did you also look to see if 'the number of nights when the bats left the colony' predicted the bat's earlier enrichment treatment? This could also be interesting to consider.

In response to the reviewer's comment, we conducted an additional analysis to test whether the proportion of nights each bat spent foraging outside the roost was predicted by its earlier environmental condition (enriched vs. impoverished). We also examined whether sex or age influenced this variable. This analysis showed no significant effect of environmental condition, sex, or age on the proportion of nights spent foraging outside the roost

[Following on point 3 in public review...]

When wishing to discuss the effect/significance of predictors overall, it is common to present the modelling results as an analysis of variance table. See, for example, the two-way anova section (p. 182) in the book Practical Regression and ANOVA using R: <https://cran.r-project.org/doc/contrib/Faraway-PRA.pdf>

I think the output of passing the model object to an "anova" yields the table that you may be looking for, where the variance accounted for by a predictor is given overall, and not just relative to the first level of all predictors. Naturally, this information can be used in combination with the information provided by the raw model output presented in the paper.

I assume you have done this analysis in R, but am not sure, as the statistical software used is not mentioned. There are several packages in R that allow users to quickly plot the graphical interaction of the parameters they use in models, which aids in interpreting results. It would be good to check results of model fitting in this manner.

Relatedly, I was unable to locate the data and code for this paper using the DOI provided. Neither searching the internet using the doi nor entering the doi on the Mendeley Data website returned the right results. I tried searching Mendeley Data using the senior author's last name, but the most recent entry does not appear to be from this paper. <https://data.mendeley.com/datasets/fr48bmnhxj/1>

We thank the reviewer for the helpful comment. The analysis was indeed conducted in MATLAB, and this has now been clarified in the manuscript. We have also revised the result tables to improve clarity and included the exact formulas used for each model. Regarding the data availability, the reviewer is correct — the dataset had not yet been published at the time of submission. It is now available at the provided DOI link.

Suggestions and questions for the present paper, grouped thematically:

[Major] Expansion and development of results: I thought there were many interesting and suggestive points in this data that could be expanded upon. I mention some of these here. While the authors of course do not need to implement all of these suggestions, I think the paper would benefit from a more substantial presentation of this rich data set:

(a) Individual differences as such are not emphasized in the paper so much, as the analyses, particularly those expressed as boxplots, are grouped. The scatter plots in Figure 1 give the richest insight into how individual behavior changes throughout the course of the experiment. I would advocate for the authors to show additional comparisons using such scatter plots (perhaps in the supplementary, if needed).

We thank the reviewer and added scatter plots to figure 2

(b) In the second paragraph of the results, the authors introduce the concept of a pareto front and that of personality archetypes (lines 101-107). I found this very interesting, but these concepts were never reiterated upon later in the results or in the discussion. In fact, at many points, I found myself curious as to how the three indoor measures of personality might be combined to form a composite measure of personality (and likewise for outdoor measures). Have you tried to combine measures into a composite and tried to measure whether this composite metric provides any additional insight into these phenomena? For example, what if you mapped the starting position of each bat as a point in a three-dimensional space, given by the three personality measures, and then evaluated their trajectory through this space with measurements taken at later trials. Could innate personality be interpreted as the starting vector in this space (measured across the two baseline trials)?

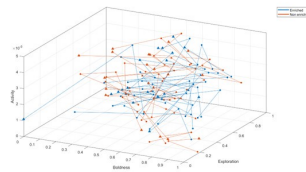
Following the reviewer's (justified) curiosity we ran a PCA analysis on the behavioral data from trials 1 and 5 and found that there is a significant correlation between the individual scores on PC1. This can be thought of as a measurement that takes both boldness and exploration into account (the weight of activity was very low). We added this information to the revised manuscript and also use this new behavioral parameter as a predisposition in the models (instead of exploration and activity).

Could environmental exposure be quantified as a warping of the trajectory through this space? Finally, could outdoor experience also be incorporated to evaluate how an individual arrives at its final measurement of personality combined with experience (trial 5)?

The paper currently tries to explain outdoors behavior given personality and not vice versa. While this is a very interesting suggestion, we feel that adding this analysis would make the premise of the paper less clear and since the paper is already somewhat complex, we prefer to leave this analysis for a future study.

Examining the 3D trajectories of the individuals through the personality space did not reveal any immediate clear pattern (triangles mark the first trial and colours depict the environmental treatment) –

Author response image 1.



Related to this point: I think the strongest part of the paper is the result showing that bats exposed to enriched environments explore farther, more often, and over larger distances than bats that were raised in an impoverished environment.

We completely agree and tried to further emphasize this

(c) While these results of the outdoor GPS tracking are very clear, I wish that more information were extracted from the tracking data, which is incredibly rich and certainly can be used to derive many interest parameters beyond those that the authors have shown here. Examples might include: distance travelled (as opposed to estimated km² or farthest point), a metric of navigational ability (how much "dead reckoning" the animal engages in). I even wonder if the areas or landmarks visited by the enriched bats might be found to be more complex, challenging, or richer by some measure.

This study was a first step, aiming to establish a connection between early exposure and outdoors foraging

We agree that there are many more analyses that can be done and indeed that ones related to navigation capabilities are missing. We are still collecting data on these bats and hope to present a more advanced analysis with a time span of years.

(d) Related to the above point: I find it very interesting that in 3 of the 4 bats for which you show exemplary movement data (Fig. 3, panels B and C), they appear to travel to the farthest distances and cover the most ground early on, and become more "conservative" in their flight paths on later evenings. This point is not explored in the discussion, nor related to earlier measurements.

During the first months of exploration, bats will occasionally perform long exploratory flights in between bouts of shorter flights where they return to nearby familiar trees. This behavior can be seen in more detail in Harten et al Science 2020. We are currently quantifying this more carefully for another study.

(e) Finally, my points about the possible strength of a composite measure of the three personality metrics is related to my concern about one of the conclusions, which is that innate personality does not have an effect on outdoor foraging behavior. I think the manner in which this was tested statistically is likely to bias the results against finding such a result given that personality metrics are used to predict outdoor behaviors in an individual manner (6 models in total, each examining a single comparison of predisposition to outdoor behavior), while both indoor personality metrics (Fig 1B) and outdoor behaviors appear to be correlated with each other (Table 5).

Are there other analyses you have performed that are not presented in the paper and that have led you to conclude that there is no relationship here?

We agree with the reviewer, that our findings do not exclude an effect of innate personality on foraging but only suggest no such effect for the parameter we measured. That said, we did expect to find an effect of boldness because this parameter has been shown to differentiate much between groups (Harten et al., 2018), and to correlate with other parameters of behavior. We were therefore surprised to find no significant effects, as we had anticipated observing some differences.

Following the reviewer's previous comment we now also tested another predisposition parameter – the PC1 score and also found that it did not explain foraging.

(f) Personality measured before and after early environmental exposure (related to point (a) above): I find it interesting that the positive correlation in boldness between baseline and post-enrichment or baseline and post-release suggests that the individuals that were the most bold remained bold (and likewise for less adventurous individuals). The correlation for activity, too, still suggests that more active individuals early in life are likely to remain very active after enrichment, even accounting for the fact that activity is confounded with age.

Perhaps you could place some emphasis on the fact that the initial variation between individuals also appears to be relatively stable over repeated trials. You might also consider measuring this directly (population variance over successive trials; relationship of population variance on indoor measures vs. outdoor measures...)

Yes – this is a main point of interest. We further emphasize that in the revised manuscript

(g) Effect of indoor behavior following early experience on outdoor behavior: You evaluate the effect of predisposition (measured on baseline trial 1) and environmental condition on measures of outdoor activity (Table 4). I wonder if you also tried using indoor behavioral measures measured on the post-enrichment trial 3 to predict outdoor foraging behavior.

Assuming that these measures are in fact reflecting a combination of predisposition and accumulated experience, then measurements at this closer time point may tell you how the combination of innate traits and early acquired experience affect behavior in the wild.

We appreciate the reviewer's insightful suggestion to test whether indoor behavior from post-enrichment Trial 3, reflecting both innate traits and experience, predicts outdoor foraging behavior. We conducted this analysis, but found that the boldness in Trial 3 did not significantly predict any of the outdoor activity measures.

(2) [Minor] Age/development: While the authors discuss the effect of their manipulations on behavioral measures, they do not much discuss the effect of age.

*I think it would be important to include at some point a mention of the developmental stages of *Rousettus*, giving labels to certain age ranges, e.g. pup, juvenile, adult, and to provide more context about the stages at which bats were tested in the discussion. Presently, age is only really mentioned as an explanation for declining activity levels, but I wonder if it might also have an influence on boldness.*

It would also be very elegant for figures where age is given in days, to additional label then with these stages.

All bats were juveniles during the trials (approximately 4 to 8 months old), so they could not be divided into distinct age groups. To assess the effect of age, it was included as a predictor

(in days) in the GLM analysis.

(3) [Major] Effect of early experience and outdoor experience on the indoor task: In the paragraph on lines 278-285, you argue that the effect of seeing earlyenriched bats exhibit more boldness in trial 5 was likely due to post-sampling bias...

I tend to disagree with this conclusion. I actually find this result both interesting and intuitive - that bats that were exposed to an enriched environment and have had experience in the wild, show much bolder activity on a familiar indoor foraging test (i.e. outside experience has made the animals bolder than before) (Fig 1, lines 159-161, Fig. S1). I did not notice this possibility mentioned in the discussion of the results.

I also do not fully understand this argument. Could you please explain further?

We accept the reviewer's comment and updated the manuscript (lines 336346) explaining the two hypotheses more clearly and arguing that it is difficult to tell them apart with the current data.

[Minor] You also say that "this difference... can be seen in Figure 2 when examining only the bats that had remained until the last trial (Figure 2A2)." Do you mean supplementary Figure S1 A2? In fact, I am entirely unclear on what data is plotted in the supplementary Figure S1 and what differentiates the two columns of figures and the two models presented in the supplementary table. Did you plot data similar to that in Figure 2, with only bats that were present for all trials, but not show this data?

There was a mistake: what was previously referred to as 2A2 is actually S2 A2.

On the right side—only among the individuals with GPS data—the change is already evident at Baseline 2, where only the bolder individuals remain. If you have suggestions for a better analysis approach, we would be happy to hear them.

Minor points

General points regarding figures:

For Figures 2 and 3A1-3 (as well as Fig. S1): Authors must show the raw data points over the box plots. It is very difficult to interpret the data and conclusions without being able to see the true distribution.

Done

For all figures showing grouped individual data, please annotate all panels or sets of boxplots with the number of bats whose data entered into each, as it is a little difficult to keep track of the changing sample sizes across experimental stages.

To enhance transparency, we have added individual data points to all boxplots, allowing visual estimation of sample sizes across experimental stages. While numerical annotations are not included on the figures, the exact number of bats contributing to each group is provided in the Methods section (Table 8), ensuring this information is readily accessible to readers. In response to the reviewer's request, we have updated all relevant figures to display individual data points within each boxplot. This addition makes it easier to track changes in sample size across different experimental stages.

Unless I've missed the reason behind differences in axis labelling across the figures, it seems that trials are not always referred to consistently. E.g. Fig. 1 labels say "Trial 1 (baseline)" and fig. 2 labels say "Baseline 1 0 days." I'm not entirely sure if these

correspond to exactly the same data. If so, perhaps the labels can be made uniform. I think the descriptive ones (Baseline 1, Postenrichment...) may be more helpful to the reader than providing the trial number (Trial 1, etc....).

Done

Figure 1:

Very good Fig. 1A and 1B.

For panels C1-3 & D, I think it would make it easier for the reader if the personality measure labels were placed at the top of each panel, e.g. "Boldness (entrance proportion)". The double axis labels are not only harder to read, they are also redundant, as the personality measure label repeats on both axes.

Done

Panel C1: For the first panel in this sequence, I think it would be elegant to include an annotation in the figure that indicates what the datapoints lying on either side of the dashed line means, i.e. "bolder after enrichment treatment" in the upper left corner, and "bolder before enrichment treatment" in the bottom right corner.

Panel C2: It appears as though many of the data points in this panel overlap, and it appears to me that the blue data points in particular are overlaid by the orange ones. I am guessing this happens because proportion values based on entrances to only 6 boxes end up giving a more "discrete" looking distribution. I wonder if you can find a way to allow all the data to be visible by, e.g., jittering the data slightly; if there is rounding being done to the proportions, perhaps don't round them so that minute differences will allow them to escape the overlap; or possibly split the panel by enrichment treatment.

Caption for C1-3: it may be helpful to mention the correlation line color scheme: "enriched (blue lines), the impoverished (orange lines)". The caption also says positive correlations were found for "both environments together," but this correlation line is not shown. Perhaps mention "(not shown)" or show line. Please rephrase the sentence "Dashed line represents the $Y=X$ line." for more transparency and clarity. I understand you mean an "equality" or "unity" line, but perhaps you can explicitly state the information that this line provides, something like e.g. "Dashed line indicates equal values measured on both trials."

We added the line for a reference, the caption was corrected

Figure 3:

Panels B1-C2: I would suggest giving these panels supertitles that indicate that B panels are enriched, C panels are impoverished, and that each panel is data from a different individual.

The legend was corrected to be more clear about the figure

General points regarding tables:

Please revisit tables for formatting and typos, particularly in Table 4. Please also revise table captions for clarity. E.g. "first exploration as predisposition" to "Exploration (Baseline 1)" or similar

Done

Supplementary Tables and Figure: these are missing captions and explanations.

The missing parts were added and corrected

Points of clarification/style:

It would seem to me more logical to present the results shown in Table 3 before those in Table 2, given that the primary in-lab manipulation is discussed with relation to Table 3, and the analysis in Table 2 is discussed rather as a limitation (though I believe this result can be expanded upon further, see above).

For the activity metric, I would suggest showing this data as actions/hour instead of actions/minute. I think it is much more intuitive to consider, for example, that a bat makes 2 actions every hour, than that it makes 0.002 actions per minute.

Done

<https://doi.org/10.7554/eLife.103220.2.sa0>